

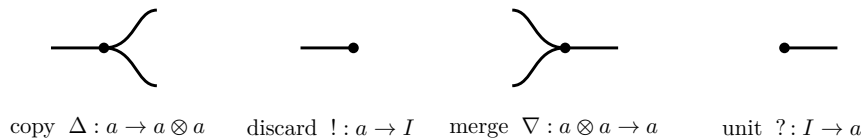
§2.124 in string diagrams — a proof from the axioms

Freyd & Scedrov, *Categories, Allegories* §2.124: $\text{Dom}(R \cap S) = 1 \cap SR^\circ$ — “a lemma we will use repeatedly”

An earlier note only **drew** the two sides and declared them equal. Here is the actual proof: the generators, the axioms they obey, two lemmas, and a step-by-step calculation in point-free (AOP) style, then drawn. Every algebraic step is tagged with the axiom it uses. The three underlying identities are proved in Lean for *arbitrary* relations — not a random sample — in `Freyd/S2_124.lean` (sorry-free, axioms `[propext, Classical.choice, Quot.sound]`).

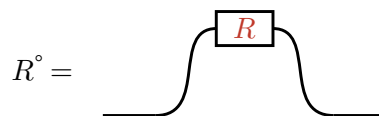
§1. What a string diagram is

An arrow of a symmetric monoidal category drawn as a circuit: white boxes are the **relations** $R, S : a \rightarrow b$; wires carry objects. Left-to-right juxtaposition is composition $;$ (the book’s $\times$ y), vertical stacking is the monoidal product $\otimes$ (the book’s $\oplus$). The identity 1 is a bare wire. An **allegory**’s operations are recovered from **one** extra structure on every object: a black-dotted **special Frobenius (co)monoid** — four generators.

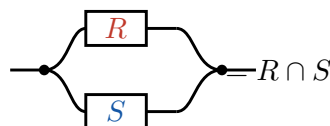


In REL these are the diagonal relations: **copy** is $x \mapsto (x, x)$, **discard** is $x \mapsto \checkmark$, **merge** = copy° , **unit** = discard° . Two derived pieces are used constantly — **converse** and **meet**:

Converse R° is **free**: bend both of R ’s wires around, so what was an input is read as an output. In REL, $x R^\circ y$ holds exactly when $y R x$.



Meet $R \cap S$ is **not** a new generator; it is the convolution $R \cap S = \Delta; (R \otimes S); \nabla$: copy the input, run R and S in parallel, then **merge**. The merge forces the two outputs to **coincide** — it demands both xRy and xSy .



The **domain** used below is $\text{Dom } R = 1 \cap R R^\circ$ (Freyd §2.122): the coreflexive $\sqsubseteq 1$ picking out the a that R relates to **something**.

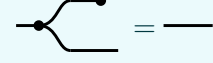
§2. The axioms of the calculus

Definition 2.1 (Special commutative Frobenius algebra (per object)).

The four dots obey these equations — the only moves a diagram may be rewritten by. σ is

the wire swap; **?** is the **unit** : a wire **created from nothing** ($? : \mathbf{I} \rightarrow \mathbf{a}$), the converse of discard **!**. The pair $(\Delta, !)$ copies and deletes; its converse pair $(\nabla, ?)$ merges and creates. Each law is drawn beside it (the monoid $(\nabla, ?)$ laws — assoc, unit, commutative — are the vertical mirrors of the first three):

counit $\Delta; (1 \otimes !) = 1$

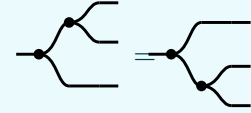


cocommutative $\Delta; \sigma = \Delta$



coassociative $\Delta; (\Delta \otimes 1) = \Delta; (1 \otimes \Delta)$

the picture re-brackets three wires for free; *coassoc* in the Lean spells the associator *assocLR* out

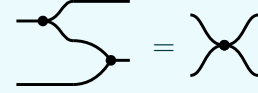


special $\Delta; \nabla = 1$ *copy-then-merge = wire*



Frobenius $(\Delta \otimes 1); (1 \otimes \nabla) = \nabla; \Delta = (1 \otimes \Delta); (\nabla \otimes 1)$

same suppression: *frobenius* in the Lean carries *assocLR* too, because *Prod* is not strictly associative



Converse $(-)^{\circ}$ is the involution reversing composition and fixing the structure:

$$(P; Q)^{\circ} = Q^{\circ}; P^{\circ}, \quad P^{\circ\circ} = P, \quad \Delta^{\circ} = \nabla, \quad !^{\circ} = ?, \quad (P \otimes Q)^{\circ} = P^{\circ} \otimes Q^{\circ},$$

whence $(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$. With $\text{cup} = ?; \Delta$ and $\text{cap} = \nabla; !$, the special and Frobenius laws give the **yanking** (snake) equations, so a wire may be straightened.

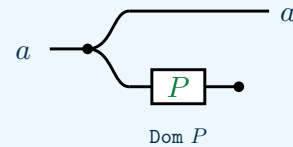
Derived — spider fusion. From the four groups above, any **connected** diagram built only from $\Delta, \nabla, !, ?$ with m inputs and n outputs equals the **single spider** $s_{(m,n)}$. This is the one power tool: a tangle of dots collapses to one node with the right number of legs.

§3. Two lemmas

Lemma 3.1 (domain = copy · run · discard).

$$\text{Dom } P = 1 \cap PP^{\circ} = \Delta; (1 \otimes P); (1 \otimes !).$$

Copy the input; run P on one copy; **discard** its output. The discard is the existential “ P relates to something”, so this is the domain coreflexive. (Both sides denote $\{(a, a) : \exists b. aPb\}$.)

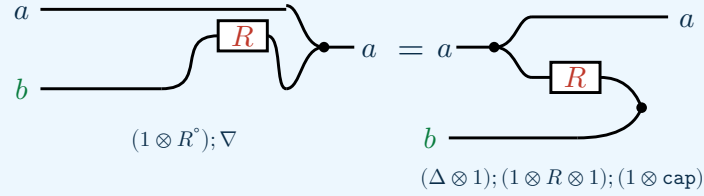


Lemma 3.2 (converse folds into a cap).

 For $R : a \rightarrow b$,

$$(1 \otimes R^\circ); \nabla = (\Delta \otimes 1); (1 \otimes R \otimes 1); (1 \otimes \text{cap}).$$

A converse R° feeding a merge equals: **copy** the surviving wire, run R forward, and **cap** its output against the incoming one — just “bend R , then yank” (converse + Frobenius). This is the move that turns $S R^\circ$ into a witness.



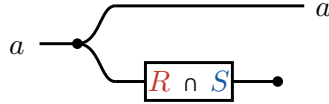
§4. The proof

Theorem 4.1.

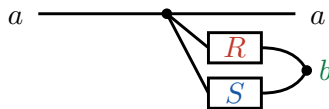
$$1 \cap SR^\circ = \text{Dom}(R \cap S).$$

Read the pictures top to bottom: $\text{Dom}(R \cap S)$ and $1 \cap S R^\circ$ both reduce to the **same** middle diagram W , so $\text{Dom}(R \cap S) = W = 1 \cap S R^\circ$. Each $=$ is one graphical move.

Dom(RnS) Lemma 1: copy a , run $R \cap S$ on one copy, discard

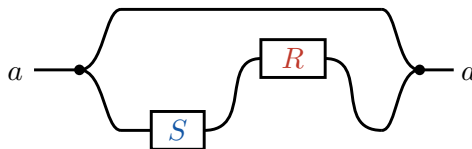


$=$ unfold $R \cap S = \Delta; (R \otimes S); \nabla$; $\nabla; ! = \text{cap}$; fuse the copies (coassoc)



the normal form $W = \Delta^3; (1 \otimes R \otimes S); (1 \otimes \text{cap})$ tap a ; run R, S cap to one b

$=$ Lemma 2: bend the R° straight



$1 \cap S R^\circ$ copy; the top wire is the 1 ; S then the bent R° ; merge

The one step with content is **Lemma 2** (bottom $=$): bending the R° in $S R^\circ$ straightens the $S-R^\circ$ loop into R beside S , both fed by the same tap, their b -outputs meeting at the

cap — the shared witness b . Everything else is unfolding n and spider fusion. No modular law appears; §2.124 is here **derived** from Frobenius + converse.

§5. Is this checked in Lean?

Yes — every boxed result above is a Lean theorem in `Freyd/S2_124.lean`, sorry-free, for arbitrary relations R, S :

in this note	in <code>Freyd/S2_124.lean</code>
Lemma 3.1 (domain = copy · run · discard)	<code>dom_cd</code>
Lemma 3.2 (converse folds into a cap)	<code>cv_merge</code>
Theorem 4.1 $1 \cap SR^\circ = \text{Dom}(RnS)$	<code>dom_inter_rel</code> — via <code>left_eq_W</code> , <code>right_eq_W</code>
the axioms of §2 (special, Frobenius, coassoc, ...)	<code>special</code> , <code>frobenius</code> , <code>coassoc</code> , <code>counit</code> , ...

One honest caveat about **how** Lean checks them. Lean works in the concrete model `REL` (`Rel A B := A → B → Prop`) and confirms each equality holds there for all relations; the §2 axioms are also proved there, so `REL` is shown to be a model of the calculus. The step-by-step diagram **rewriting** above is the human “why”; a fully point-free Lean replay of it would additionally need the associator/unitor coherence lemmas (`Prod` is not strictly associative). The `1nPP° → Δ; (1⊗P); !` collapse of Lemma 1 needs one extra relational law beyond bare Frobenius — **adequacy** ($\Delta; (R \otimes R); \nabla = R$) — stated explicitly in the file. No modular law is used anywhere.

The abstract, allegory-level §2.124 that **does** use the modular law is `dom_inter` in `Freyd/S2_1.lean`; its two halves are the two moves this diagram fuses into one:

<code>dom_inter</code> step (allegory / AOP)	string-diagram move
$1 \cap SR^\circ \sqsubseteq 1 \cap (RnS)R^\circ$ — <code>modular_le S R[°] 1</code>	Lemma 2: bend R° , so R hangs beside S
$= 1 \cap R(R^\circ n S^\circ)$ — a coreflexive equals its reciprocal	read the tapped loop either way (converse of a coreflexive = itself)
$\sqsubseteq (RnS)(R^\circ n S^\circ) = \text{Dom}(RnS)$ — <code>modular_le R (R[°]nS[°]) 1</code>	coassoc: the two prongs share one tap Δ^3 and one cap
the outer 1 kept at each step (<code>le_inter</code> with $\sqsubseteq 1$)	the straight diagonal wire — the spider only taps it, never deletes it